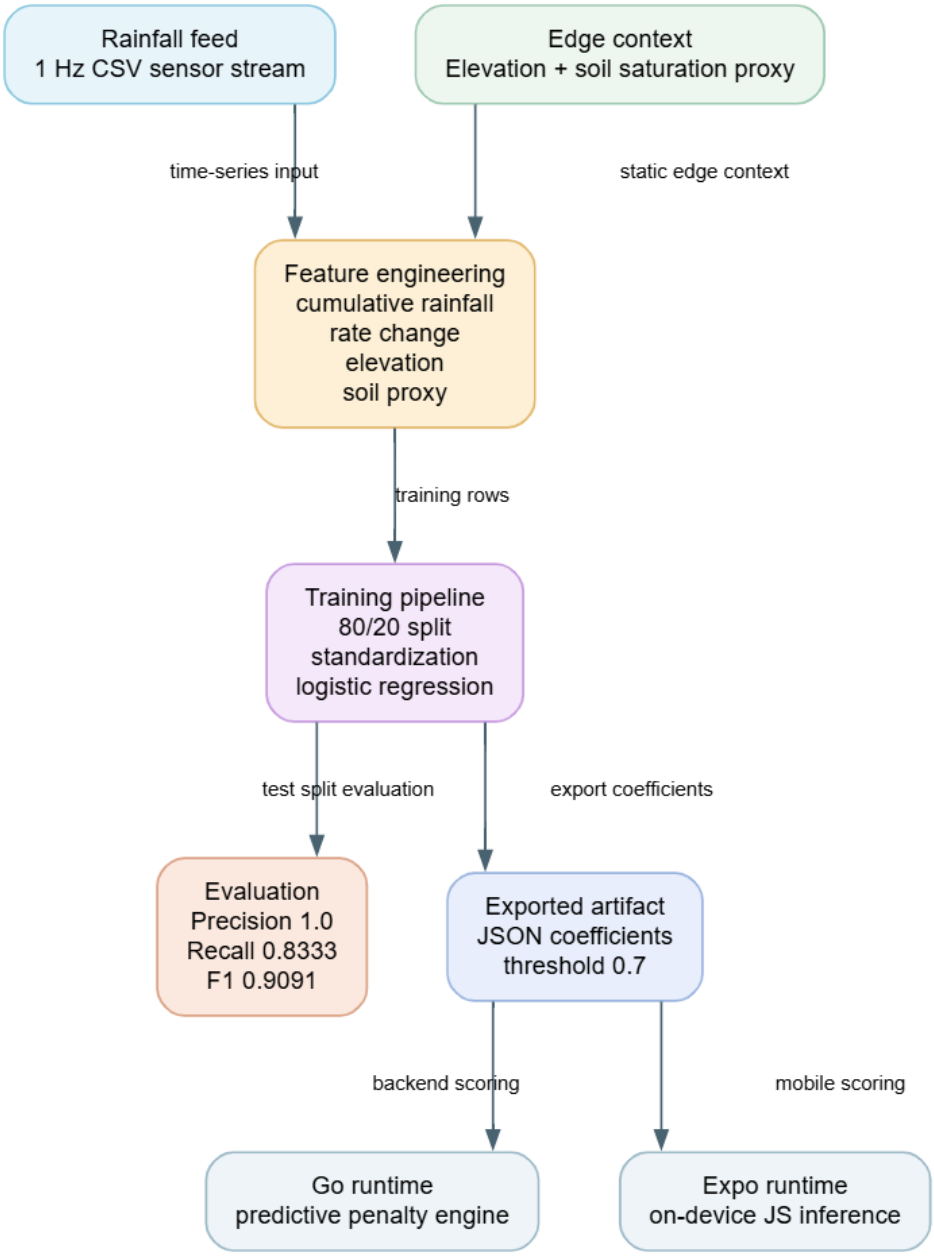
Route Decay Model Card

Huntrix Delta — HackFusion 2026 | Logistic regression for 2-hour edge impassability prediction

Summary: The model predicts whether a route edge will become impassable within the next **2 hours**. Features are derived from the rainfall feed and edge context, then exported as JSON coefficients for both the Go backend and Expo mobile app.

Methodology Diagram



Evaluation Results

Metric	Precision	Recall	F1	Accuracy
Value	1.0	0.8333	0.9091	0.98

Confusion Counts	TP	FP	TN	FN
Value	5	0	44	1

Notes

- Training split: **80/20**
- Threshold: **0.7**
- Features: `cumulative_rainfall_mm`, `rainfall_rate_change`, `elevation_m`, `soil_saturation_proxy`
- Main limitation: labels are synthetic, so real-world confidence is lower than the demo metrics suggest.

Primary artifacts: `ml/artifacts/route_decay_model.json`, `ml/artifacts/route_decay_metrics.json`, `docs/model-card.md`